CBSE Class 10 Information Technology

Unit 5: Green Skills – I

Session 1: Environment and Us

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1. Introduction

The environment is everything that surrounds us. It includes living and non-living things that affect our life. Humans, animals, plants, air, water, land, and other natural elements together form the environment.

A healthy environment is essential for the survival and well-being of all living organisms.

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2. What is Environment?

Environment refers to all the surroundings, conditions, and influences in which living organisms exist and interact.

Examples:

- Air
- Water
- Soil
- Plants
- Animals
- Human beings

Definition:

Environment is the sum total of all living and non-living things around us.

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3. What Makes the Natural Environment?

The **Natural Environment** consists of all naturally occurring things that are not created by humans.

Components of Natural Environment:

1. Land (Lithosphere)
2. Water (Hydrosphere)
3. Air (Atmosphere)
4. Living organisms (Biosphere)

Examples:

- Mountains

- Rivers
- Forests
- Oceans
- Animals
- Plants

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4. Types of Environment (Sub-categories of Natural Environment)

The natural environment can be divided into two main categories:

A. Living Environment (Biotic Environment)

It includes all living organisms.

Examples:

- Humans
- Animals
- Birds
- Insects
- Plants
- Microorganisms

Characteristics:

- Can grow
- Can reproduce
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- Need food and water
- Respond to surroundings

B. Non-Living Environment (Abiotic Environment)

It includes all non-living things.

Examples:

- Air
- Water
- Soil
- Sunlight

- Rocks
- Minerals

Characteristics:

- Do not grow
- Do not reproduce
- Support living organisms

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5. Types of Resources

A resource is anything that can be used to satisfy human needs.

Examples:

- Water
- Forests
- Minerals
- Coal
- Petroleum

Resources are mainly classified as:

A. Natural Resources

Resources obtained directly from nature are called natural resources.

Examples:

- Water
- Air
- Soil
- Forests
- Minerals

Importance:

- Support life on Earth
- Provide food, energy, and raw materials

B. Renewable and Non-Renewable Resources

1. Renewable Resources

Resources that can be replenished naturally in a short period of time.

Examples:

- Sunlight
- Wind Energy
- Water
- Forests

Features:

- Can be used repeatedly
- Environment-friendly

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2. Non-Renewable Resources

Resources that take millions of years to form and cannot be replaced quickly.

Examples:

- Coal
- Petroleum
- Natural Gas
- Minerals

Features:

- Limited in quantity
- May get exhausted

C. Exhaustible and Non-Exhaustible Resources

1. Exhaustible Resources

Resources that can be depleted due to excessive use.

Examples:

- Coal
- Petroleum
- Natural Gas
- Forests

Features:

- Limited availability

- Need careful use

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2. Non-Exhaustible Resources

Resources that are available in unlimited quantity and are not likely to be exhausted.

Examples:

- Sunlight
- Air
- Wind

Features:

- Available continuously
- Sustainable source of energy

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6. Relationship Between Society and Environment

Society and environment are closely connected.

How Society Depends on Environment:

- Food comes from plants and animals.
- Water is needed for drinking and farming.
- Natural resources provide energy and raw materials.
- Forests maintain ecological balance.

How Society Affects Environment:

- Industrialization
- Urbanization
- Deforestation
- Pollution

Conclusion:

Humans depend on the environment for survival, and therefore must protect it.

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7. Factors Causing Environmental Imbalance

Environmental imbalance occurs when natural systems are disturbed.

Major Causes:

1. Deforestation

- Cutting down trees excessively.
- Causes soil erosion and loss of biodiversity.

2. Pollution

- Air pollution
- Water pollution
- Soil pollution
- Noise pollution

3. Industrialization

- Release of harmful gases and waste.

4. Population Growth

- Increased demand for resources.

5. Overexploitation of Resources

- Excessive use of coal, petroleum, forests, etc.

6. Climate Change

- Rise in global temperature.
- Changes in weather patterns.

Important Terms

Term	Meaning
Environment	Surroundings in which living organisms exist
Natural Environment	Environment created by nature
Biotic Environment	Living components of environment
Abiotic Environment	Non-living components of environment
Natural Resources	Resources obtained from nature
Renewable Resources	Resources that can be replenished naturally
Non-Renewable Resources	Resources that cannot be replaced quickly
Exhaustible Resources	Resources that can be depleted
Non-Exhaustible Resources	Resources available in unlimited quantity

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Very Short Answer Questions

1. What is environment?
    - Environment is the surroundings in which living organisms live.
2. Name two biotic components.
    - Plants and animals.
3. Give two examples of abiotic components.
    - Air and water.
4. What are renewable resources?
    - Resources that can be replenished naturally.
5. Give one example of a non-renewable resource.
    - Coal.
6. What is deforestation?
    - Large-scale cutting of trees.
7. Name two non-exhaustible resources.
    - Sunlight and wind.

Session 2: Our Influence on the Natural Environment

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1. Introduction

Human beings depend on nature for food, water, air, energy, and other resources. With the increase in population, industries, and technology, humans are using natural resources at a much faster rate. This has caused serious environmental problems such as pollution, global warming, deforestation, and climate change.

To ensure a sustainable future, it is important to use natural resources wisely and protect the environment.

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2. Human Impact on the Environment

Human impact refers to the positive or negative effects of human activities on the natural environment.

Positive Impacts

- Planting trees (Afforestation)
- Recycling waste
- Saving electricity and water
- Using renewable energy (solar and wind)
- Wildlife conservation

Negative Impacts

- Deforestation
- Pollution
- Illegal mining
- Industrialization
- Overuse of natural resources
- Burning fossil fuels

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3. Factors Causing Interference in the Environment

Human activities disturb the natural balance of the environment.

Major Factors:

- Rapid population growth
- Urbanization

- Industrial development
- Deforestation
- Excessive use of natural resources
- Pollution
- Mining activities
- Burning fossil fuels

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4. Population Explosion

Population explosion means a rapid increase in the number of people living on Earth.

As the population increases, the demand for food, water, land, housing, and energy also increases.

Effects of Population Explosion

A. More Consumption of Land

- Forests and farmland are converted into cities, roads, and buildings.
- Wildlife loses its natural habitat.

B. More Forests Being Cut (Deforestation)

- Trees are cut to build houses, industries, and roads.
- Reduces oxygen production.
- Increases carbon dioxide in the atmosphere.
- Causes soil erosion and loss of biodiversity.

C. More Production of Carbon Dioxide (CO_2)

- More vehicles and industries burn fossil fuels.
- More CO_2 is released into the atmosphere.
- Leads to global warming.

D. More Consumption of Water

- Increased demand for drinking, farming, and industries.
- Causes water scarcity in many areas.
- Reduces groundwater levels.

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5. Pollution

Pollution is the contamination of the environment by harmful substances.

It affects human health, animals, plants, and the ecosystem.

Causes of Pollution

A. More Waste Production

- Increasing population generates large amounts of solid waste.
- Plastic waste pollutes land and water.

B. Fuel Usage

- Burning petrol, diesel, coal, and natural gas releases harmful gases.
- Causes air pollution.

C. Poor Waste Management

- Improper disposal of garbage.
- Open dumping and burning of waste.
- Pollutes soil, water, and air.

Types of Pollution

1. Air Pollution

Cause: Smoke from vehicles, factories, and burning fuels.

Effects:

- Respiratory diseases
- Global warming
- Acid rain

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2. Water Pollution

Cause: Industrial waste, sewage, and chemicals entering water bodies.

Effects:

- Unsafe drinking water
- Death of aquatic life
- Spread of diseases

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3. Soil Pollution

Cause: Plastic waste, pesticides, chemicals, and industrial waste.

Effects:

- Loss of soil fertility
- Harm to plants and animals

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4. Noise Pollution

Cause: Loudspeakers, vehicles, factories, and construction work.

Effects:

- Hearing problems
- Stress
- Sleep disturbances

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6. Global Warming

Global warming is the gradual increase in the Earth's average temperature due to the excessive release of greenhouse gases like carbon dioxide (CO_2).

Main Causes

- Burning fossil fuels
- Deforestation
- Industrial emissions
- Vehicle pollution

Effects

- Melting glaciers
- Rising sea levels
- Extreme weather conditions
- Heat waves
- Climate change
- Loss of biodiversity

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7. Illegal Mining

Illegal mining is the extraction of minerals without legal permission or by violating environmental laws.

Effects of Illegal Mining

A. Loss of Habitat

- Forests are destroyed.

- Animals lose their homes.
- Biodiversity decreases.

B. Causes Pollution

- Dust causes air pollution.
- Chemicals pollute rivers and soil.
- Noise from mining machines disturbs wildlife.

C. Disturbs Water Levels

- Lowers groundwater levels.
- Pollutes underground and surface water.
- Affects agriculture and drinking water.

D. Climate Change

- Deforestation during mining reduces carbon absorption.
- Increases greenhouse gas emissions.
- Contributes to global warming.

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8. Climate Change

Climate change refers to long-term changes in the Earth's temperature and weather patterns.

Causes

- Global warming
- Deforestation
- Pollution
- Burning fossil fuels
- Industrial activities

Effects

- Floods
- Droughts
- Cyclones
- Heat waves
- Irregular rainfall
- Loss of biodiversity
- Reduced agricultural production

Important Terms

Term	Meaning
Human Impact	Effect of human activities on the environment
Population Explosion	Rapid increase in human population
Deforestation	Large-scale cutting of forests
Pollution	Contamination of the environment
Global Warming	Increase in Earth's average temperature
Illegal Mining	Mining without legal permission
Climate Change	Long-term changes in weather and climate patterns

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Very Short Answer Questions

1. What is population explosion?
Rapid increase in the human population.

2. What is pollution?
Contamination of the environment by harmful substances.

3. What is global warming?
The gradual increase in the Earth's average temperature due to greenhouse gases.

4. What is deforestation?
The large-scale cutting of trees and forests.

5. What is illegal mining?
Mining carried out without legal permission or by violating environmental laws.

6. Name two causes of air pollution.

- Vehicle emissions
- Factory smoke

7. Name two effects of global warming.

- Melting glaciers
- Rising sea levels

8. Give two effects of illegal mining.

- Loss of wildlife habitat
- Water pollution

Session 3: Natural Resource Conservation

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1. Introduction

Natural resources such as air, water, forests, soil, minerals, and wildlife are essential for life on Earth. Due to increasing population and excessive use of these resources, many natural resources are getting depleted.

Natural Resource Conservation means protecting, managing, and using natural resources wisely so that they are available for present and future generations.

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2. Saving and Conserving the Natural Environment

Conservation means using natural resources carefully without wasting them.

Importance of Conservation

- Protects biodiversity.
- Maintains ecological balance.
- Prevents pollution.
- Saves resources for future generations.
- Supports sustainable development.

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3. Reduce Carbon Footprint

What is Carbon Footprint?

A **carbon footprint** is the total amount of greenhouse gases (mainly carbon dioxide - CO_2) released into the atmosphere due to human activities.

Examples include using electricity, driving vehicles, burning fuels, and industrial activities.

Ways to Reduce Carbon Footprint

A. Reduce Electricity Consumption

- Switch off lights, fans, and appliances when not in use.
- Use LED bulbs.
- Use energy-efficient electrical appliances.
- Use solar energy whenever possible.

B. Use Greener Transportation

- Walk or ride a bicycle.
- Use public transport.
- Share vehicles (carpooling).
- Use electric or hybrid vehicles.

C. Change Eating Habits

- Avoid wasting food.
- Eat more locally produced food.
- Reduce excessive consumption of processed food.
- Choose eco-friendly and sustainable food options.

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4. The 3Rs – Reduce, Reuse and Recycle

The **3Rs** help minimize waste and conserve natural resources.

A. Reduce

Reduce the use of unnecessary products and resources.

Examples

- Save electricity.
- Save water.
- Avoid single-use plastic.
- Print only when necessary.

B. Reuse

Use an item again instead of throwing it away.

Examples

- Reuse shopping bags.
- Refill water bottles.
- Reuse glass jars and containers.

C. Recycle

Convert waste materials into new useful products.

Examples

- Recycling paper.
- Recycling plastic.
- Recycling metal.
- Recycling glass.

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5. Soil Conservation

What is Soil Conservation?

Soil conservation means protecting soil from erosion, pollution, and loss of fertility.

Healthy soil is essential for agriculture and food production.

Causes of Soil Degradation

A. Land Overuse

- Excessive farming without allowing the land to recover.
- Reduces soil fertility.

B. Slash and Burn Farming

A farming method in which forests are cut down and burned to create farmland.

Effects

- Destroys forests.
- Causes soil erosion.
- Reduces soil fertility.
- Increases air pollution.

C. Chemical Contamination

Excessive use of fertilizers and pesticides pollutes the soil.

Effects

- Reduces soil fertility.
- Harms microorganisms.
- Pollutes groundwater.

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6. Soil Conservation Techniques

A. No-Till Farming

A farming method where the soil is not ploughed before planting.

Benefits

- Reduces soil erosion.
- Improves soil health.
- Conserves moisture.

B. Crop Rotation

Growing different crops on the same land in different seasons.

Benefits

- Maintains soil fertility.
- Controls pests.
- Improves crop production.

C. Terrace Farming

Steps or terraces are made on hill slopes for farming.

Benefits

- Prevents soil erosion.
- Conserves water.
- Suitable for hilly regions.

D. Windbreaks

Rows of trees or shrubs planted around farms.

Benefits

- Reduce wind speed.
- Prevent soil erosion.
- Protect crops.

E. Earthworms

Earthworms naturally improve soil quality.

Benefits

- Increase soil fertility.
- Improve soil structure.
- Help in decomposition of organic matter.

F. Making River Dams

Dams help control water flow and reduce soil erosion.

Benefits

- Prevent floods.

- Store water for irrigation.
- Reduce soil erosion.

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7. Water Conservation

What is Water Conservation?

Water conservation means using water wisely and preventing wastage.

Methods of Water Conservation

- Rainwater harvesting.
- Repair leaking taps.
- Turn off taps when not in use.
- Use drip irrigation.
- Reuse water whenever possible.
- Build dams and reservoirs.
- Protect rivers and lakes.

Importance

- Prevents water scarcity.
- Ensures water for future generations.
- Supports agriculture.
- Maintains groundwater levels.

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8. Energy Conservation

What is Energy Conservation?

Energy conservation means reducing unnecessary use of energy.

Ways to Save Energy

- Switch off electrical appliances when not in use.
- Use LED bulbs.
- Use energy-efficient appliances.
- Use renewable energy sources like solar and wind energy.

- Use public transport.
- Maintain electrical equipment properly.

Benefits

- Saves money.
- Reduces pollution.
- Conserves fossil fuels.
- Reduces greenhouse gas emissions.

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9. Forest Conservation

What is Forest Conservation?

Forest conservation means protecting forests and using forest resources sustainably.

Methods of Forest Conservation

A. Plant More Trees

Planting more trees increases forest cover and improves air quality.

B. Regulate the Cutting of Trees

- Stop illegal logging.
- Cut only mature trees.
- Plant new trees after cutting.

C. Forest Protection

- Protect forests from illegal activities.
- Prevent encroachment.
- Protect wildlife habitats.

D. Fire Suppression Techniques

Measures taken to prevent and control forest fires.

Examples:

- Fire lines

- Early warning systems
- Firefighting equipment
- Public awareness

E. Development of Sanctuaries and National Parks

Protected areas help conserve wildlife and forests.

Benefits

- Protect endangered species.
- Preserve biodiversity.
- Promote eco-tourism.

F. Reforestation and Afforestation

Reforestation

Planting trees again in areas where forests have been cut down.

Afforestation

Planting trees on land where there were no forests earlier.

Difference Between Reforestation and Afforestation

Reforestation	Afforestation
Trees are replanted in previously forested areas.	Trees are planted on barren or non-forest land.
Restores lost forests.	Creates new forests.

Important Terms

Term	Meaning
Conservation	Careful use and protection of natural resources
Carbon Footprint	Total greenhouse gases released by human activities
Reduce	Use fewer resources
Reuse	Use an item again
Recycle	Convert waste into useful products

Term	Meaning
Soil Conservation	Protection of soil from erosion and pollution
Crop Rotation	Growing different crops on the same land in different seasons
Terrace Farming	Farming on step-like terraces on hills
Windbreaks	Rows of trees planted to reduce wind erosion
Water Conservation	Wise use of water resources
Energy Conservation	Saving energy by reducing unnecessary use
Reforestation	Replanting trees in deforested areas
Afforestation	Planting trees on land that was not previously forested

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Very Short Answer Questions

1. What is conservation?
The careful use and protection of natural resources.

2. What is a carbon footprint?
The total amount of greenhouse gases released by human activities.

3. What are the 3Rs?
Reduce, Reuse, and Recycle.

4. What is soil conservation?
Protecting soil from erosion and loss of fertility.

5. What is crop rotation?
Growing different crops on the same land in different seasons.

6. What is terrace farming?
Farming on step-like terraces built on hill slopes.

7. What is reforestation?
Replanting trees in areas where forests were cut down.

8. What is afforestation?
Planting trees on barren or non-forest land.

9. Name two methods of water conservation.

- Rainwater harvesting
- Repairing leaking taps

10. Name two methods of energy conservation.

- Using LED bulbs
- Switching off appliances when not in use

Session 4: Green Economy and Green Skills

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1. Introduction

The rapid use of natural resources, pollution, and climate change have created serious environmental problems. To protect the environment while ensuring economic growth and social well-being, the concept of a **Green Economy** has been introduced.

A green economy promotes sustainable development by reducing pollution, conserving natural resources, and creating environmentally friendly jobs.

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2. Definition of Green Economy

A **Green Economy** is an economy that improves human well-being and social equality while reducing environmental risks and conserving natural resources.

It focuses on **low carbon emissions, efficient use of resources, and sustainable development**.

Three Pillars of Green Economy

A. Environmental

- Protects nature and biodiversity.
- Reduces pollution.
- Conserves natural resources.
- Promotes renewable energy.

B. Social

- Improves quality of life.
- Creates employment opportunities.
- Promotes equality and better public health.
- Ensures sustainable living for all.

C. Economic

- Encourages sustainable economic growth.
- Creates green businesses and industries.
- Increases resource efficiency.
- Supports long-term development.

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3. Importance of Green Economy

A green economy is important because it:

- Conserves natural resources.
- Reduces pollution.
- Lowers greenhouse gas emissions.
- Promotes sustainable development.
- Creates green jobs.
- Improves public health.
- Protects biodiversity.
- Helps fight climate change.
- Supports future generations.

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4. Elements of Green Economy

The major elements of a green economy are:

A. Generation and Use of Renewable Energy

Renewable energy comes from natural sources that can be replenished.

Examples

- Solar energy
- Wind energy
- Hydropower
- Biomass energy
- Geothermal energy

Benefits

- Reduces pollution.
- Saves fossil fuels.
- Produces clean energy.
- Reduces carbon emissions.

B. Energy Efficiency

Energy efficiency means using less energy to perform the same task.

Examples

- Using LED bulbs.
- Energy-efficient appliances.
- Proper insulation in buildings.
- Switching off unused electrical devices.

Benefits

- Saves electricity.
- Reduces energy costs.
- Decreases pollution.
- Conserves natural resources.

C. Waste Minimization and Management

Waste minimization means reducing the amount of waste produced.

Waste management involves collecting, treating, recycling, and safely disposing of waste.

Methods

- Reduce
- Reuse
- Recycle
- Composting
- Proper segregation of waste

Benefits

- Reduces pollution.
- Conserves resources.
- Protects public health.
- Keeps the environment clean.

D. Preservation and Sustainable Use of Natural Resources

Natural resources should be used wisely without harming future generations.

Examples

- Saving water.
- Protecting forests.
- Conserving wildlife.
- Sustainable farming.
- Responsible mining.

Benefits

- Maintains ecological balance.
- Prevents resource depletion.
- Supports biodiversity.

E. Green Job Creation

Green jobs help protect the environment while providing employment.

Examples

- Solar panel technician
- Wind turbine technician
- Environmental engineer
- Organic farmer

- Waste management worker
- Forest officer

Benefits

- Reduces pollution.
- Promotes clean technologies.
- Supports sustainable development.
- Creates employment opportunities.

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5. Green Skills

Green Skills are the knowledge, abilities, values, and attitudes required to protect the environment and use natural resources efficiently.

Green skills help individuals work in environmentally friendly ways and support sustainable development.

A. Green Skills Required for Managing Resource Efficiency

Resource efficiency means using natural resources carefully with minimum wastage.

Skills Required

- Efficient use of water and electricity.
- Waste reduction.
- Recycling practices.
- Sustainable production methods.
- Energy-saving techniques.

B. Green Skills Required for a Low Carbon Footprint

These skills help reduce greenhouse gas emissions.

Skills Required

- Using renewable energy.
- Energy conservation.
- Eco-friendly transportation.
- Planting trees.
- Reducing fuel consumption.
- Following the 3Rs (Reduce, Reuse, Recycle).

C. Green Skills Required for Climate Resilience

Climate resilience means the ability to prepare for, respond to, and recover from climate-related problems.

Skills Required

- Disaster preparedness.
- Water conservation.
- Sustainable agriculture.
- Environmental awareness.
- Climate adaptation practices.

D. Green Skills Required for Managing Natural Assets

Natural assets include forests, rivers, wildlife, minerals, soil, and water resources.

Skills Required

- Forest conservation.
- Wildlife protection.
- Soil conservation.
- Water resource management.
- Biodiversity conservation.
- Sustainable use of natural resources.

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6. Green Jobs

A **Green Job** is a job that helps protect the environment and conserve natural resources while supporting economic growth.

Green jobs reduce pollution, save energy, and promote sustainable development.

Examples of Green Jobs

- Environmental Scientist
- Forest Officer
- Wildlife Conservationist
- Solar Panel Technician
- Wind Energy Technician
- Waste Management Officer

- Recycling Plant Worker
- Organic Farmer
- Water Conservation Specialist
- Environmental Engineer
- Energy Auditor
- Eco-Tourism Guide

Difference Between Green Skills and Green Jobs

Green Skills	Green Jobs
Knowledge and abilities to protect the environment.	Employment that helps protect the environment.
Learned through education and training.	Careers that use green skills.
Help in sustainable living.	Support a green economy.

Important Terms

Term	Meaning
Green Economy	Economy that promotes environmental protection, social well-being, and sustainable economic growth
Renewable Energy	Energy from natural sources that can be replenished
Energy Efficiency	Using less energy to perform the same work
Waste Management	Collection, treatment, recycling, and safe disposal of waste
Natural Resources	Resources obtained from nature
Green Skills	Skills that support environmental protection and sustainability
Green Jobs	Jobs that conserve natural resources and reduce pollution
Climate Resilience	Ability to adapt to and recover from climate-related challenges
Carbon Footprint	Total greenhouse gases released by human activities

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Very Short Answer Questions

1. What is a green economy?
An economy that promotes environmental protection, social well-being, and sustainable economic growth.

2. Name the three pillars of a green economy.

- Environmental
- Social
- Economic

3. What are green skills?
Skills that help protect the environment and use natural resources efficiently.

4. What is renewable energy?
Energy obtained from natural sources that are replenished naturally.

5. Give two examples of renewable energy.

- Solar energy
- Wind energy

6. What is energy efficiency?
Using less energy to perform the same task.

7. What is a green job?
A job that helps protect the environment and conserve natural resources.

8. Name two green jobs.

- Solar Panel Technician
- Forest Officer

9. What is waste management?
The process of collecting, treating, recycling, and safely disposing of waste.

10. What is climate resilience?
The ability to prepare for, adapt to, and recover from climate-related changes and disasters.
